

Volume III. Seventh Section.
Transformation equations.
Sections 74–76.

§ 74. The form of the Schlaefli modular equations for a prime degree.

The form which we found in the preceding paragraph for the relations existing between

$$u = f(\omega), \quad v = f\left(\frac{c + \partial\omega}{a}\right) \quad (1)$$

can, at least when the degree of transformation is a prime number p , easily be brought under a general law. Since in what follows much depends on this form, we shall enter here into somewhat more detail about it.

The determination of the numbers r, s according to (11) and (15) of the preceding paragraph depends on the behavior of p with respect to the modulus 24; and since we may exclude the case $p = 3$, we have the following cases:

$p \pmod{24}$	r	s
1	1	12
5	3	2
7	4	3
11	6	1
13	1	6
17	3	4
19	2	3
23	12	1,

(2)

so that $r + s$ is always odd, and $p + 1$ is divisible by $2r$, $p - 1$ by $2s$. Accordingly

$$A = \left(\frac{u}{v}\right)^r + \left(\frac{v}{u}\right)^r, \quad B = (uv)^s + \left(\frac{2}{p}\right) \frac{2^s}{(uv)^s}. \quad (3)$$

Now we know that the $p - 1$ quantities v are the roots of an irreducible transformation equation

$$\Phi_p(v, u) = v^{p+1} + U_1 v^p + \cdots + U_{p+1} = 0, \quad (4)$$

in which the coefficients $U_1, \dots, U_{p+1}$ are rational functions of u .

We conclude at once that they are integral rational functions of u . For first, when $u = \infty$, all roots of (4) become infinite, as is seen by putting $\omega = i\infty$, hence letting $q = 0$. Secondly, according to (9) of the preceding paragraph, by the interchange

$$\left(u, \frac{\sqrt{2}}{u}\right)$$

the totality of the roots v goes over into

$$\left(\frac{2}{p}\right) \frac{\sqrt{2}}{v}.$$

From this it follows that for $u = 0$ all roots v go over into zero, so that none of them becomes infinite. Hence one concludes not only that $U_1, U_2, \dots, U_{p+1}$ are integral functions of u , but also that each of them must contain the factor u .

We conclude next, exactly as for the invariant equation (§ 69, 2., 3. and 6.), that

$$\Phi_p(v, u) = \Phi_p(u, v), \quad (5)$$

and that

$$\Phi_p(v, u) = (v^p - u)(v - u^p) + p \sum_{1,p}^{h,k} c_{h,k} u^h v^k, \quad (6)$$

where $c_{h,k} = c_{k,h}$ are integers.

We shall represent this function in the form

$$\Phi_p(v, u) = v^{p+1} + u^{p+1} + \sum_{1,p}^{h,k} a_{h,k} u^h v^k, \quad (7)$$

where therefore $a_{h,k} = a_{k,h}$ are likewise integers, whose divisibility by p is now no longer relevant, and in particular $a_{p,1} = -1$. If in the equation

$$\Phi_p[f(p\omega), f(\omega)] = 0$$

one replaces ω by $\omega + 2$, then, because of irreducibility and on the basis of the relation

$$f(\omega + 2) = e^{-\pi i/12} f(\omega),$$

it follows that in (7) not all terms occur, but only those for which the exponents h, k satisfy the congruence

$$hp + k \equiv p + 1 \pmod{24}. \quad (8)$$

Now we know one more property of the functions $\Phi_p(v, u)$, which results from the interchange (9) of the preceding paragraph already used, where $\rho = 1$ is to be put; if, for abbreviation, we put

$$\varepsilon = \left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8},$$

this property can be written

$$\Phi_p(v, u) = \left(\frac{uv}{\sqrt{2}}\right)^{p+1} \Phi_p\left(\frac{\varepsilon\sqrt{2}}{v}, \frac{\sqrt{2}}{u}\right). \quad (9)$$

From this one obtains the relations

$$\varepsilon^h 2^{(h+k-p-1)/2} a_{p+1-h, p+1-k} = a_{h,k}, \quad a_{h,k} = a_{k,h}. \quad (10)$$

If, therefore, we combine in (7) the terms with equal coefficients $a_{h,k}$, we may restrict ourselves to the assumption that $h \geq k$, $h + k \geq p + 1$; and, apart from the two terms v^{p+1} and u^{p+1} , Φ_p appears as an aggregate of terms of the following two forms (taking into account also that, by (8), $\varepsilon^h = \varepsilon^k$):

$$\begin{aligned} & v^h u^k + v^k u^h - \varepsilon^h 2^{(h+k-p-1)/2} (u^{p+1-h} v^{p+1-k} + u^{p+1-k} v^{p+1-h}) \\ &= (uv)^{(p+1)/2} \left[\left(\frac{v}{u}\right)^{(h-k)/2} + \left(\frac{u}{v}\right)^{(h-k)/2} \right] \left[(uv)^{(h+k-p-1)/2} + \frac{\varepsilon^h 2^{(h+k-p-1)/2}}{(uv)^{(h+k-p-1)/2}} \right], \end{aligned} \quad (11)$$

and

$$\begin{aligned} & v^h u^h + \varepsilon^h 2^{h-(p+1)/2} (uv)^{p+1-h} \\ &= (uv)^{(p+1)/2} \left[(uv)^{h-(p+1)/2} + \frac{\varepsilon^h 2^{h-(p+1)/2}}{(uv)^{h-(p+1)/2}} \right]. \end{aligned} \quad (12)$$

The coefficients of these terms in Φ_p are integers.

Now (8) gives

$$h - k \equiv (h - 1)(p + 1), \quad h + k - p - 1 \equiv -h(p - 1) \pmod{24},$$

and hence $h - k$ is divisible by $2r$, and $h + k - p - 1$ (where h may also be equal to k) by $2s$ [§ 74, (2)].

If, therefore, we put

$$\frac{h - k}{2} = r\alpha, \quad \frac{h + k - p - 1}{2} = s\beta,$$

then α and β are positive integers which lie between the bounds

$$0 \leq \alpha < \frac{p + 1}{2r}, \quad 0 \leq \beta \leq \frac{p - 1}{2s}, \quad (13)$$

and in addition the congruence following from (8), $-h(p - 1) \equiv 2s\beta \pmod{24}$, shows that at least in the cases where $\varepsilon = -1$ one has $h \equiv \beta \pmod{2}$ [§ 74, (2)], and hence always $\varepsilon^h = \varepsilon^\beta$.

Thus (11) becomes

$$(uv)^{(p+1)/2} \left[\left(\frac{v}{u} \right)^{r\alpha} + \left(\frac{u}{v} \right)^{r\alpha} \right] \left[(uv)^{s\beta} + \frac{\varepsilon^\beta 2^{s\beta}}{(uv)^{s\beta}} \right],$$

and (12) becomes

$$(uv)^{(p+1)/2} \left[(uv)^{s\beta} + \frac{\varepsilon^\beta 2^{s\beta}}{(uv)^{s\beta}} \right].$$

Now the well-known formulae hold, when x, y are arbitrary quantities and

$$x + \frac{\gamma}{x} = y$$

is put:

$$\begin{aligned} x^2 + \frac{\gamma^2}{x^2} &= y^2 - 2\gamma, & x^3 + \frac{\gamma^3}{x^3} &= y^3 - 3\gamma y, \dots, \\ \left(x^n + \frac{\gamma^n}{x^n} \right) \left(x + \frac{\gamma}{x} \right) &= x^{n+1} + \frac{\gamma^{n+1}}{x^{n+1}} + \gamma \left(x^{n-1} + \frac{\gamma^{n-1}}{x^{n-1}} \right), \end{aligned}$$

from which one recognizes, by induction from n to $n + 1$, that

$$x^n + \frac{\gamma^n}{x^n}$$

can, for arbitrary n , be represented as an integral rational function of degree n in y , which, if γ is an integer, has integral coefficients and leading coefficient 1.

The two quantities

$$\left(\frac{v}{u} \right)^{r\alpha} + \left(\frac{u}{v} \right)^{r\alpha} \quad \text{and} \quad (uv)^{s\beta} + \frac{\varepsilon^\beta 2^{s\beta}}{(uv)^{s\beta}}$$

can therefore be represented in this way as integral rational functions of A and B of degrees α and β , and we find, if we also separate off the term corresponding to $\beta = (p - 1)/(2s)$, which has coefficient -1 , and denote integral coefficients by $c_{\alpha, \beta}$:

$$\frac{\Phi_p(u, v)}{(uv)^{(p+1)/2}} = A^{(p+1)/(2r)} - B^{(p-1)/(2s)} + \sum c_{\alpha, \beta} A^\alpha B^\beta, \quad (14)$$

where α and β are bound by the limits

$$0 \leq \alpha < \frac{p + 1}{2r}, \quad 0 \leq \beta < \frac{p - 1}{2s}.$$

This is the form which in the preceding paragraph we gave to the modular equations up to $p = 19$.

It remains to mention that the consideration carried out in § 69, 4., 5. for the invariant equation in the case of composite transformation degree applies unchanged also to the Schlaefli modular equations. Hence we may conclude that all these equations have rational numerical coefficients.

§ 75. The irrational forms of the modular equations.

By applying the same procedure, much simpler forms can be given to the transformation equations. The equation forms with which we shall now be concerned contain the three functions f, f_1, f_2 simultaneously; but since, by § 34, (11), (12), two of these functions can be expressed by the third, two of them can be eliminated, and one can thus arrive at the equations of the preceding paragraph. If one substitutes for f_1, f_2 their expressions through f , or for the three functions f, f_1, f_2 their expressions through k^2 , radical signs occur; from this the name of these equations is explained.

We put, as above,

$$\begin{aligned} u &= f(\omega), & v &= f\left(\frac{c + \partial\omega}{a}\right), \\ u_1 &= f_1(\omega), & v_1 &= \left(\frac{2}{a}\right) f_1\left(\frac{c + \partial\omega}{a}\right), \\ u_2 &= f_2(\omega), & v_2 &= \left(\frac{2}{\partial}\right) f_2\left(\frac{c + \partial\omega}{a}\right), \\ uu_1u_2 &= \sqrt{2}, & vv_1v_2 &= \left(\frac{2}{n}\right) \sqrt{2}, \end{aligned} \tag{1}$$

and obtain the following associated interchanges:

$$\begin{array}{c|ccc} \omega & uv & u_1v_1 & u_2v_2 \\ -1/\omega & \varrho uv & \varrho u_2v_2 & \varrho u_1v_1 \\ \omega + 1 & e^{-(n+1)\pi i/24} \sigma u_1v_1 & e^{-(n+1)\pi i/24} \sigma uv & e^{(n+1)\pi i/12} \sigma u_2v_2, \end{array} \tag{2}$$

where ϱ, σ are the third roots of unity defined above in § 73, (4).

We distinguish three different cases according to the behavior of n modulo 8.

1. $n + 1 \equiv 0 \pmod{8}$.

We put

$$\begin{aligned} 2A &= uv + (-1)^{(n+1)/8} (u_1v_1 + u_2v_2), \\ B &= uvu_1v_1 + uvu_2v_2 + (-1)^{(n+1)/8} u_1v_1u_2v_2 \\ &= \frac{2}{u_1v_1} + \frac{2}{u_2v_2} + (-1)^{(n+1)/8} \frac{2}{uv}, \end{aligned} \tag{3}$$

so that the associated interchanges become

$$\begin{array}{c|cc} \omega & A & B \\ -1/\omega & \varrho A & \varrho^2 B \\ \omega + 1 & e^{\pi i(n+1)/12} \sigma A & e^{-\pi i(n+1)/12} \sigma^2 B. \end{array} \tag{4}$$

A product of the form $A^h B^k$ therefore takes, under the two interchanges

$$\left(\omega, -\frac{1}{\omega}\right), \quad (\omega, \omega + 1),$$

the factors

$$\varrho^{h-k}, \quad e^{(h-k)(n+1)\pi i/12} \sigma^{h-k},$$

which are equal to 1 when $n + 1$ is divisible by 3. We therefore now form, with numerical coefficients $M_{h,k}$, functions of the form

$$\Phi_{a,c,\partial} = \sum M_{h,k} A^h B^k, \tag{5}$$

where, if $n \equiv 0$ or $1 \pmod{3}$, h and k may assume only integral values whose differences $h - k$ leave the same residue on division by 3, while if $n \equiv -1 \pmod{3}$ they are not subject to this restriction. Such a function itself, or at least its third power, therefore satisfies a transformation equation of the first stage. It also remains finite for all finite values of $j(\omega)$, and consequently is an integral

algebraic function of $j(\omega)$. [The transformation of the second order $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ need not be brought in here, since the totality of the $\Phi_{a,c,\partial}$ remains unchanged already under the substitution $(\omega, \omega + 1)$, not only under $(\omega, \omega + 2)$. Compare the remark at the beginning of § 73.]

If, therefore, we determine the constants in $\Phi_{a,c,\partial}$ so that no negative powers of q occur in the expansions of these functions, then all the $\Phi_{a,c,\partial}$ must be equal to one and the same constant.

To reach this goal it is sufficient here also, if n has no square divisor, that $\Phi_{1,0,n}$, and if n is a prime number, that $\Phi_{1,0,n}$, remain finite for $q = 0$.

In these calculations we use the expansions

$$\begin{aligned} uv &= q^{-(n+1)/24} \prod (1 + q^{n(2h+1)})(1 + q^{2h+1}), \\ u_1 v_1 &= q^{-(n+1)/24} \prod (1 - q^{n(2h+1)})(1 - q^{2h+1}), \\ u_2 v_2 &= 2q^{(n+1)/12} \prod (1 + q^{2hn})(1 + q^{2h}), \end{aligned} \quad (6)$$

from which, by expansion in powers of q ,

$$\begin{aligned} uv &= q^{-(n+1)/24} (1 + q + q^3 + q^4 + q^5 + q^6 - q^7 + 2q^8 \cdots), \\ u_1 v_1 &= q^{-(n+1)/24} (1 - q - q^3 + q^4 - q^5 + q^6 - q^7 + 2q^8 \cdots), \\ u_2 v_2 &= 2q^{(n+1)/12} (1 + q^2 + q^4 + 2q^6 + 2q^8 + \cdots). \end{aligned} \quad (7)$$

The latter formulae are correct for $n > 7$; for $n = 3, 5, 7$ the terms from q^3, q^5, q^7 on are to be modified. For $n = 7, n = 23$ these expressions show that A itself remains finite for $q = 0$, whence for these cases the equations follow:

$$n = 7, \quad A = 0, \quad n = 23, \quad A = 1. \quad (8)$$

By a simple calculation one further finds:

$$\begin{aligned} n = 31, \quad (A^2 - B)^2 - A &= 0, \\ n = 47, \quad A^2 - A - B &= 2, \\ n = 71, \quad A^3 - 4A^2 + 2A - B &= 1. \end{aligned} \quad (9)$$

The calculation is not quite so simple for composite n . For example, for $n = 15$ one must take account of the condition of finiteness for $q = 0$ not only for $\Phi_{1,0,15}$, but also for $\Phi_{3,0,5}$. These two, however, are sufficient. One obtains:

$$n = 15, \quad A^3 - AB + 1 = 0. \quad (10)$$

2. If $n \equiv 3 \pmod{8}$, the same conclusions are to be drawn if we put

$$\begin{aligned} 4A &= u^2 v^2 - u_1^2 v_1^2 - u_2^2 v_2^2, \\ B &= u^2 v^2 u_1^2 v_1^2 + u^2 v^2 u_2^2 v_2^2 - u_1^2 v_1^2 u_2^2 v_2^2, \end{aligned} \quad (11)$$

for which one obtains from (2) the associated interchanges

$$\begin{array}{c|cc} \omega & A & B \\ -1/\omega & \varrho^2 A & \varrho B \\ \omega + 1 & e^{(n+1)\pi i/6} \sigma^2 A & e^{-(n+1)\pi i/6} \sigma B, \end{array} \quad (12)$$

and from this, in the same way, one derives the equations

$$n = 3, \quad A = 0, \quad n = 11, \quad A = 1, \quad n = 19, \quad A^5 - 7A^2 - B = 0. \quad (13)$$

3. If $n \equiv 1 \pmod{4}$, the same procedure can be followed with the functions

$$\begin{aligned} 8A &= u^4 v^4 - u_1^4 v_1^4 - u_2^4 v_2^4, \\ B &= u^4 v^4 u_1^4 v_1^4 + u^4 v^4 u_2^4 v_2^4 - u_1^4 v_1^4 u_2^4 v_2^4, \end{aligned} \quad (14)$$

for which one obtains the associated interchanges

$$\begin{array}{c|cc} \omega & A & B \\ -1/\omega & \varrho A & \varrho^2 B \\ \omega + 1 & e^{(n+1)\pi i/3} \sigma A & e^{-(n+1)\pi i/3} \sigma^2 B. \end{array} \quad (15)$$

Only the first case, $n = 5$, leads here to a simple result:

$$n = 5, \quad A = 1. \quad (16)$$

For $n = 5$ an even simpler form of the transformation equation can be obtained. If we apply the substitutions $(\omega, -1/\omega)$ and $(\omega, \omega + 1)$ to the three functions

$$w = \frac{u_1^2 v_2^2 + u_2^2 v_1^2}{uv}, \quad w_1 = \frac{u_2^2 v^2 - v_2^2 u^2}{u_1 v_1}, \quad w_2 = \frac{u_1^2 v^2 - v_1^2 u^2}{u_2 v_2}, \quad (17)$$

then, under the assumption $n = 5 \pmod{8}$, the associated interchanges become

$$\begin{array}{c|ccc} \omega & w & w_1 & w_2 \\ -1/\omega & w & w_2 & w_1 \\ \omega + 1 & e^{-\pi i(n-5)/24} w_1 & e^{-\pi i(n-5)/24} w & e^{\pi i(n-5)/12} w_2. \end{array}$$

Thus, if one sets

$$\begin{aligned} A &= w^2 + w_1^2 + w_2^2, \\ B &= w^2 w_1^2 + w^2 w_2^2 + w_1^2 w_2^2, \\ C &= w w_1 w_2, \end{aligned}$$

one obtains

$$\begin{array}{c|ccc} \omega & A & B & C \\ -1/\omega & A & B & C \\ \omega + 1 & e^{-\pi i(n-5)/12} A & e^{-\pi i(n-5)/6} B & C, \end{array}$$

so that two of these functions can be used just as A and B above. But for $n = 5$ it appears that no negative powers occur in the expansions of w, w_1, w_2 in powers of q , and therefore that A, B, C , and consequently also w, w_1, w_2 themselves, are constant.

Thus the modular equation for $n = 5$ can be put in each of the three forms

$$\begin{aligned} u_1^2 v_2^2 + u_2^2 v_1^2 &= 2uv, \\ u_2^2 v^2 - v_2^2 u^2 &= 2u_1 v_1, \\ u_1^2 v^2 - v_1^2 u^2 &= 2u_2 v_2. \end{aligned} \quad (18)$$

These equations can also be derived from the equation given by Jacobi (Fund. art. 30, collected works, vol. 1, p. 123).

We close these considerations by deriving, in the simplest cases, the Jacobian modular equations from these irrational forms.

We put [cf. § 54, (3)]

$$\frac{u_2}{u} = x = \sqrt[4]{k'}, \quad \frac{u_1}{u} = x' = \sqrt[4]{k''}, \quad \frac{v_2}{v} = y = \sqrt[4]{\lambda'}, \quad \frac{v_1}{v} = y' = \sqrt[4]{\lambda''}, \quad (19a)$$

and eliminate the quantities u and v by means of the relations

$$x^8 + x'^8 = 1, \quad xx' = \frac{\sqrt{2}}{u^3}, \quad y^8 + y'^8 = 1, \quad yy' = \frac{\sqrt{2}}{v^3}. \quad (19)$$

For $n = 3$ one obtains from (13)

$$x^2y^2 + x'^2y'^2 = 1, \quad (20)$$

a form of the modular equation which is due to Legendre. From it, by inserting for x', y' the values from (19),

$$x^8 + y^8 = 4x^2y^2 - 6x^4y^4 + 4x^6y^6,$$

or

$$(x^4 - y^4)^2 = 4(xy - x^3y^3)^2,$$

and then extracting the root and determining the sign by $q = 0$, one obtains

$$x^4 - y^4 = 2xy - 2x^3y^3, \quad (n = 3), \quad (21)$$

which, apart from the sign of $\sqrt[4]{\lambda}$ not explained more fully there, agrees with § 10, (12).

For $n = 5$ the second equation (18) gives

$$(x^2 - y^2)^3 = 4xyx'^4y'^4,$$

and hence, by squaring,

$$(x^2 - y^2)^6 = 16x^2y^2(1 - x^8 - y^8 + x^8y^8),$$

which is easily brought into the form

$$(x^2 - y^2)^2[(x^2 - y^2)^2 + 8x^2y^2]^2 = 16x^2y^2(1 - x^4y^4)^2.$$

Extracting the root gives, as above,

$$x^6 - y^6 - 4xy(1 - x^4y^4) + 5x^2y^2(x^2 - y^2) = 0 \quad (n = 5). \quad (22)$$

For $n = 7$, without extracting roots, one obtains from (8):

$$xy + x'y' = 1, \quad (23)$$

and from this

$$\begin{aligned} x^8 + y^8 - 8xy(1 + x^6y^6) + 28x^2y^2(1 + x^4y^4) \\ - 56x^3y^3(1 + x^2y^2) + 70x^4y^4 = 0 \quad (n = 7). \end{aligned} \quad (24)$$

§ 76. Composite transformation degrees.

If the transformation degree is a composite number, then one can set up still simpler transformation equations than those obtained by the path of the preceding paragraph.

We carry out these considerations here only in the simplest cases.

Let the odd transformation degree n be decomposed into two factors

$$n = n'n'', \quad (1)$$

which are relatively prime to one another.

To each transformation of degree n of the form

$$\begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix} \quad (2)$$

there can then be assigned one and only one transformation of the degrees n', n'' :

$$\begin{pmatrix} a' & 0 \\ c' & \partial' \end{pmatrix}, \quad \begin{pmatrix} a'' & 0 \\ c'' & \partial'' \end{pmatrix}, \quad (3)$$

determined by the following conditions:

$$a = a'a'', \quad \partial = \partial'\partial'', \quad \partial''c' \equiv c \pmod{a'}, \quad \partial'c'' \equiv c \pmod{a''}, \quad (4)$$

and conversely, from every pair of transformations of the form (3), according to (4), one and only one transformation (2) follows. For by (4) ∂', ∂'' are first determined as the greatest common divisors of ∂ with n' and n'' ; then c' is determined modulo a' , and c'' modulo a'' , from the two last congruences (4).

It is now chiefly important to show that the congruences (4) remain preserved when the transformations (2) and (3), according to § 69, (8) to (12), are transformed by the two linear transformations

$$(\omega, \omega + 1), \quad \left(\omega, -\frac{1}{\omega}\right).$$

According to the formulae just mentioned we set:

$$\begin{aligned} \begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ c_1 & \partial \end{pmatrix}, \\ \begin{pmatrix} a' & 0 \\ c' & \partial' \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ \lambda' & 1 \end{pmatrix} \begin{pmatrix} a' & 0 \\ c'_1 & \partial' \end{pmatrix}, \\ \begin{pmatrix} a'' & 0 \\ c'' & \partial'' \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ \lambda'' & 1 \end{pmatrix} \begin{pmatrix} a'' & 0 \\ c''_1 & \partial'' \end{pmatrix}, \end{aligned} \quad (5)$$

$$\begin{aligned} \begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} &= \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} a_2 & 0 \\ c_2 & \partial_2 \end{pmatrix}, \\ \begin{pmatrix} a' & 0 \\ c' & \partial' \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} &= \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} \begin{pmatrix} a'_2 & 0 \\ c'_2 & \partial'_2 \end{pmatrix}, \\ \begin{pmatrix} a'' & 0 \\ c'' & \partial'' \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} &= \begin{pmatrix} \alpha'' & \beta'' \\ \gamma'' & \delta'' \end{pmatrix} \begin{pmatrix} a''_2 & 0 \\ c''_2 & \partial''_2 \end{pmatrix}. \end{aligned} \quad (6)$$

First, by § 69, (9), one has

$$c_1 \equiv c + \partial \pmod{a}, \quad c'_1 \equiv c' + \partial' \pmod{a'}, \quad c''_1 \equiv c'' + \partial'' \pmod{a''},$$

whence, by (4),

$$\begin{aligned} \partial''c'_1 &\equiv \partial''c' + \partial \equiv c_1 \pmod{a'}, \\ \partial'c''_1 &\equiv \partial'c'' + \partial \equiv c_1 \pmod{a''}, \end{aligned}$$

in agreement with the congruences (4).

For the composition (6), according to § 69, (12), $\partial_2, \partial'_2, \partial''_2$ are the greatest common divisors of a, c ; of a', c' ; and of a'', c'' ; since however ∂'' is relatively prime to a' and $\partial''c' \equiv c \pmod{a'}$, ∂'_2 is also the greatest common divisor of a' and c , and for the same reasons ∂''_2 is the greatest common divisor of a'' and c . It follows that $\partial'_2, \partial''_2$ are relatively prime:

$$\partial_2 = \partial'_2\partial''_2, \quad a_2 = a'_2a''_2.$$

Further, by § 69, (11), (12), (13),

$$\partial_2\alpha = -c\gamma - a, \quad c_2 = -\partial\alpha,$$

$$\begin{aligned}\partial'_2 \alpha' &= \alpha' c' - \partial' a', & c'_2 &= -\partial' \alpha', \\ \partial''_2 \alpha'' &= \alpha'' c'' - \partial'' a'', & c''_2 &= -\partial'' \alpha'',\end{aligned}$$

and hence, by (4),

$$\partial'_2 c''_2 - c_2 \partial'_2 \equiv \alpha \alpha' (\partial c' - c \partial') \equiv 0 \pmod{a'},$$

therefore also

$$\partial'_2 c''_2 - c_2 \partial'_2 \equiv 0 \pmod{a'_2},$$

in agreement with (4); and in the same way

$$\partial''_2 c'_2 - c_2 \partial''_2 \equiv 0 \pmod{a''_2}.$$

Thus the proof is completed that the association of the transformations expressed by (4),

$$\begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix}, \quad \begin{pmatrix} a' & 0 \\ c' & \partial' \end{pmatrix}, \quad \begin{pmatrix} a'' & 0 \\ c'' & \partial'' \end{pmatrix},$$

is not disturbed by applying any linear transformation to ω . Since here we suppose n odd, we can always assume c, c', c'' divisible by 16; and if n , and consequently also n', n'' , are not divisible by 3, then c, c', c'' can also be assumed divisible by 3. But if n is divisible by 3, then one of the two factors n', n'' , say n'' , will be divisible by 3, the other n' not. Then c may still be assumed divisible by 3, but not c' and c'' . In this case the dependence of c'' on c is to be determined more precisely by the congruence

$$\partial' c'' \equiv c \pmod{3a''}. \quad (7)$$

A solution of this congruence can always be derived from a solution of the congruence (4), $\partial' c'' \equiv c \pmod{a''}$, by adding to c'' a multiple of a'' .

The congruence (7) then has, by § 69, (9) to (13), the consequence

$$\lambda \equiv \lambda' \lambda'' \pmod{3}, \quad (8)$$

$$\alpha \equiv \alpha'' \partial'_2, \quad \beta \equiv \beta'' a'_2, \quad \gamma \equiv \gamma'' a'_2, \quad \delta \equiv \delta'' \partial'_2 \pmod{3}. \quad (9)$$

Let now $v', v'_1, v'_2; v'', v''_1, v''_2$ have the same meaning for the numbers n', n'' that v, v_1, v_2 had in § 73, (3) for the number n , namely

$$\begin{aligned}v' &= f\left(\frac{c' + \partial' \omega}{a'}\right), & v'' &= f\left(\frac{c'' + \partial'' \omega}{a''}\right), \\ v'_1 &= \left(\frac{2}{a'}\right) f_1\left(\frac{c' + \partial' \omega}{a'}\right), & v''_1 &= \left(\frac{2}{a''}\right) f_1\left(\frac{c'' + \partial'' \omega}{a''}\right), \\ v'_2 &= \left(\frac{2}{\partial'}\right) f_2\left(\frac{c' + \partial' \omega}{a'}\right), & v''_2 &= \left(\frac{2}{\partial''}\right) f_2\left(\frac{c'' + \partial'' \omega}{a''}\right).\end{aligned} \quad (10)$$

We apply the interchange table (4), § 73, to these functions. Since n' is in all circumstances not divisible by 3, the cubic roots of unity ϱ', σ' are to be set equal to 1, while by the congruences (8), (9)

$$\varrho'' = \varrho^{n'}, \quad \sigma'' = \sigma^{n'} \quad (11)$$

results. Thus we obtain the following associated interchanges:

ω	v'	v'_1	v'_2
$-1/\omega$	v'	v'_2	v'_1
$\omega + 1$	$e^{-n'\pi i/24} v'_1$	$e^{-n'\pi i/24} v'$	$e^{n'\pi i/12} v'_2$
ω	v''	v''_1	v''_2
$-1/\omega$	$\varrho^{n'} v''$	$\varrho^{n'} v''_2$	$\varrho^{n'} v''_1$
$\omega + 1$	$\sigma^{n'} e^{-n''\pi i/24} v''_1$	$\sigma^{n'} e^{-n''\pi i/24} v''$	$\sigma^{n'} e^{n''\pi i/12} v''_2$

(12)

which hold together with the interchanges (4), § 73.

We now distinguish two cases.

1. If

$$(n' + 1)(n'' + 1) = 8\mu \equiv 0 \pmod{8}, \quad (13)$$

then we put

$$U = uv'v'', \quad U_1 = u_1v'_1v''_1, \quad U_2 = u_2v'_2v''_2, \quad (14)$$

$$\begin{aligned} 2A &= U + (-1)^\mu(U_1 + U_2), \\ B &= UU_1 + UU_2 + (-1)^\mu U_1 U_2 \\ &= \frac{4}{U_1} + \frac{4}{U_2} + (-1)^\mu \frac{4}{U}. \end{aligned} \quad (15)$$

From (12) and § 73, (4), we then obtain the following associated interchanges:

$$\begin{array}{c|cc} \omega & A & B \\ -1/\omega & \varrho^{n'+1}A & \varrho^{-(n'+1)}B \\ \omega + 1 & \sigma^{n'+1}e^{2\mu\pi i/3}A & \sigma^{-(n'+1)}e^{-2\mu\pi i/3}B. \end{array} \quad (16)$$

We apply our principle for deriving modular equations to these functions, and add the following remarks.

The third roots of unity occurring in (16) are equal to 1 if either n is not divisible by 3 and μ is divisible by 3, or n'' is divisible by 3 and n' leaves the residue 2. In these cases every rational function of A and B is a root of a transformation equation. In the other cases this property belongs to the cube of such a rational function of A and B in which the differences of the exponents of all terms are congruent to one another modulo 3.

If these rational functions are whole functions, and if in addition all their values are finite for $q = 0$, then they must be equal to a constant, and thereby we obtain transformation equations.

If n', n'' are prime numbers, then it is sufficient here also, as in § 72, that the negative powers of q disappear in the expansion of such a function in increasing powers of q in the one principal case, namely the case in which

$$\begin{pmatrix} a & 0 \\ c & \partial \end{pmatrix}, \quad \begin{pmatrix} a' & 0 \\ c' & \partial' \end{pmatrix}, \quad \begin{pmatrix} a'' & 0 \\ c'' & \partial'' \end{pmatrix}$$

are equal to

$$\begin{pmatrix} 1 & 0 \\ 0 & n \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & n' \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & n'' \end{pmatrix},$$

that is,

$$U = f(\omega)f(n'\omega)f(n''\omega)f(n'n''\omega);$$

for from the expansion for this one case the expansions for the remaining cases can be derived by replacing ω by

$$\frac{\omega}{n}, \quad \frac{\omega}{n'}, \quad \frac{\omega}{n''}, \quad (17)$$

and then also increasing ω by whole numbers, whereby no negative powers of q can be newly introduced.

For carrying out the calculation one uses the expansions

$$\begin{aligned} U &= q^{-(n'+1)(n''+1)/24} \prod (1 + q^{2h-1})(1 + q^{(2h-1)n'})(1 + q^{(2h-1)n''})(1 + q^{(2h-1)n}), \\ U_1 &= q^{-(n'+1)(n''+1)/24} \prod (1 - q^{2h-1})(1 - q^{(2h-1)n'})(1 - q^{(2h-1)n''})(1 - q^{(2h-1)n}), \\ U_2 &= 4q^{(n'+1)(n''+1)/12} \prod (1 + q^{2h})(1 + q^{2hn'})(1 + q^{2hn''})(1 + q^{2hn}), \end{aligned} \quad (18)$$

which, in the individual cases, give the power expansions of A, B , from which the negative powers of q are to be eliminated. In this way one calculates very simply the following equations:

$$\begin{aligned}
 n = 15, \quad A &= 1, \\
 n = 21, \quad (A^2 - B)^2 - A &= 0, \\
 n = 33, \quad A^2 - B - A &= 4, \\
 n = 35, \quad A^2 - B - A &= 2, \\
 n = 55, \quad A^3 - B - 4A^2 - A + 4 &= 0.
 \end{aligned} \tag{19}$$

2. If

$$(n' - 1)(n'' - 1) = 8\mu \equiv 0 \pmod{8},$$

then we put

$$\begin{aligned}
 A &= \frac{uv}{v'v''} + (-1)^\mu \left(\frac{u_1v_1}{v'_1v''_1} + \frac{u_2v_2}{v'_2v''_2} \right), \\
 B &= \frac{v'v''}{uv} + (-1)^\mu \left(\frac{v'_1v''_1}{u_1v_1} + \frac{v'_2v''_2}{u_2v_2} \right),
 \end{aligned} \tag{20}$$

and obtain from (12) the associated interchanges

$$\begin{array}{c|cc}
 \omega & A & B \\
 -1/\omega & \varrho^{1-n'} A & \varrho^{n'-1} B \\
 \omega + 1 & \sigma^{1-n'} e^{2\mu\pi i/3} A & \sigma^{n'-1} e^{-2\mu\pi i/3} B.
 \end{array} \tag{21}$$

Therefore we can apply the same procedure as above, if we add the restriction that only symmetric functions of A and B , that is, rational functions of AB and $A + B$, are used; for only under this assumption do all remaining values of such a function follow from one of its values by the interchanges (17). One easily calculates the following examples:

$$\begin{aligned}
 n = 15, \quad AB + 1 &= 0, \\
 n = 35, \quad 2(A + B) - AB &= 5, \\
 n = 39, \quad 2(A + B) - AB &= 3.
 \end{aligned} \tag{22}$$

The calculation presents no insurmountable difficulties even in still more complicated cases. Thus, in the paper in *Acta mathematica*, vol. II, p. 359, for the case $n = 105$, I communicated the following formula:

$$\begin{aligned}
 A &= \sum \frac{f(3\omega)f(5\omega)f(7\omega)f(105\omega)}{f(\omega)f(15\omega)f(21\omega)f(35\omega)}, \\
 B &= \sum \frac{f(\omega)f(15\omega)f(21\omega)f(35\omega)}{f(3\omega)f(5\omega)f(7\omega)f(105\omega)}, \\
 A^2 + B^2 - 4(A + B)^3 + 10AB(A + B) \\
 &\quad + 4(A + B)^2 + 10AB + 14(A + B) + 5 = 0,
 \end{aligned}$$

where the sums $\sum$ in A and B refer to the three functions f, f_1, f_2 .